

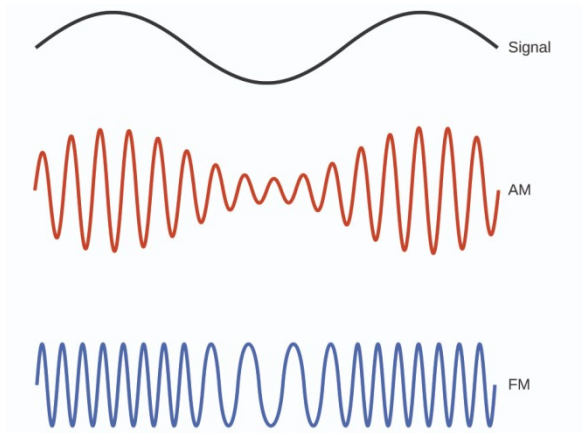
ECE 3317

Applied Electromagnetic Waves

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Fall 2025

Notes 9

Transmission Lines (Frequency Domain)



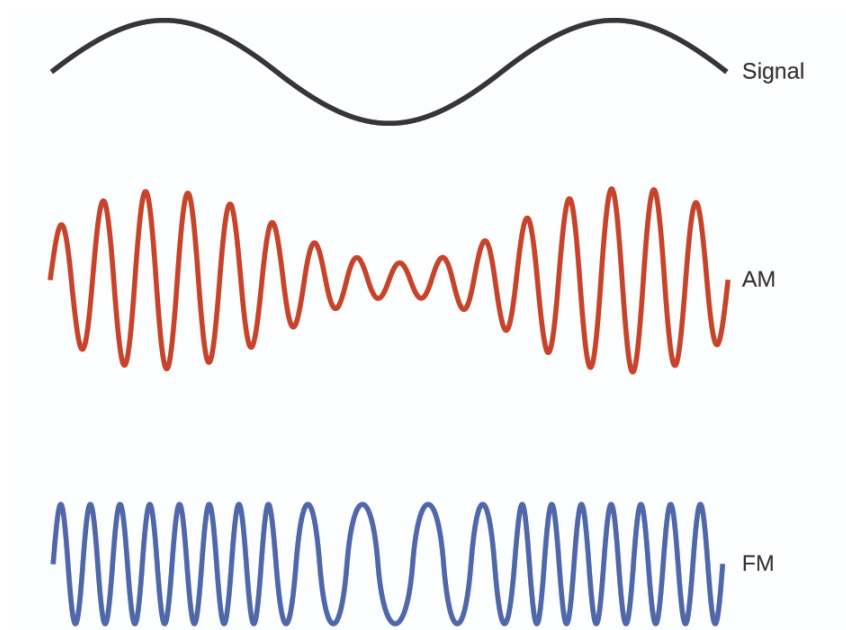
Frequency Domain

Why is the frequency domain important?

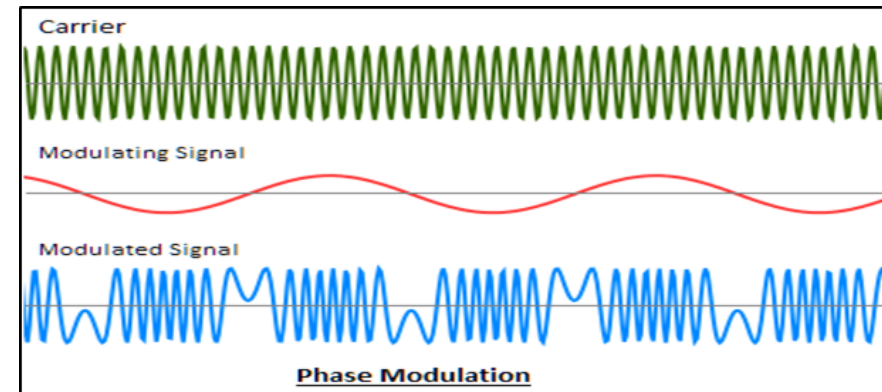
Reason # 1

❖ Most communication systems use a sinusoidal signal (which may be modulated).

(Some systems, like Ethernet, communicate in “baseband”, meaning that there is no carrier.)



↑
Examples:
100BASET, 10GBASET



Frequency Domain (cont.)

Why is the frequency domain important?

Reason # 2

- ❖ A solution in the frequency-domain allows to solve for an arbitrary time-varying signal on a lossy line (by using the Fourier transform method).

$$\begin{aligned}\tilde{v}(\omega) &= \int_{-\infty}^{\infty} v(t) e^{-j\omega t} dt \\ v(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{v}(\omega) e^{j\omega t} d\omega\end{aligned}$$

Fourier-transform pair

For a physically-realizable (real-valued) signal, we can also write

$$v(t) = \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left\{ \tilde{v}(\omega) e^{j\omega t} \right\} d\omega$$

(since $\tilde{v}(-\omega) = \tilde{v}^*(\omega)$)



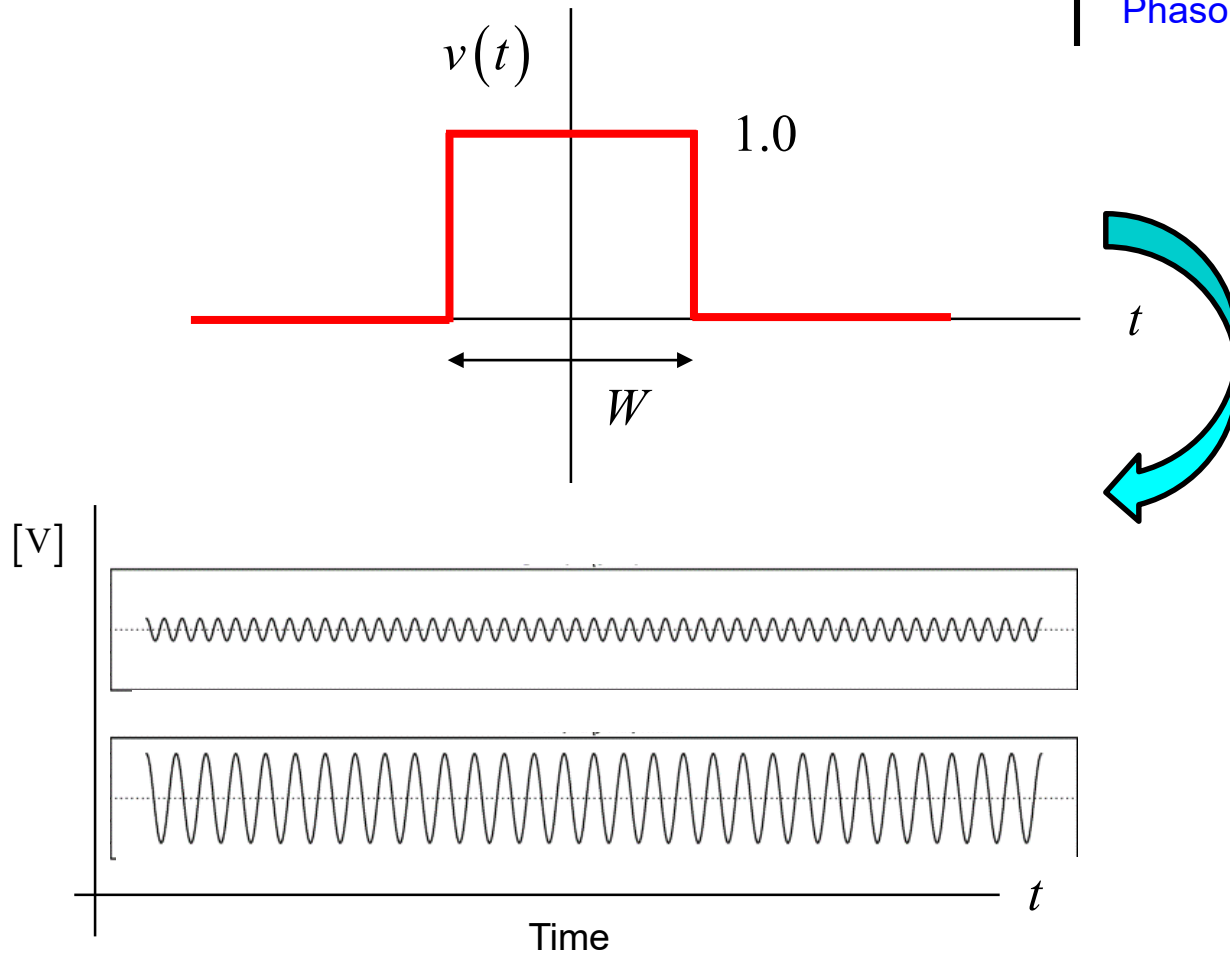
Jean-Baptiste-Joseph Fourier

Frequency Domain (cont.)

$$v(t) = \int_0^{\infty} \operatorname{Re} \left\{ \left(\frac{1}{\pi} \tilde{v}(\omega) d\omega \right) e^{j\omega t} \right\}$$

A collection of phasor-domain signals!

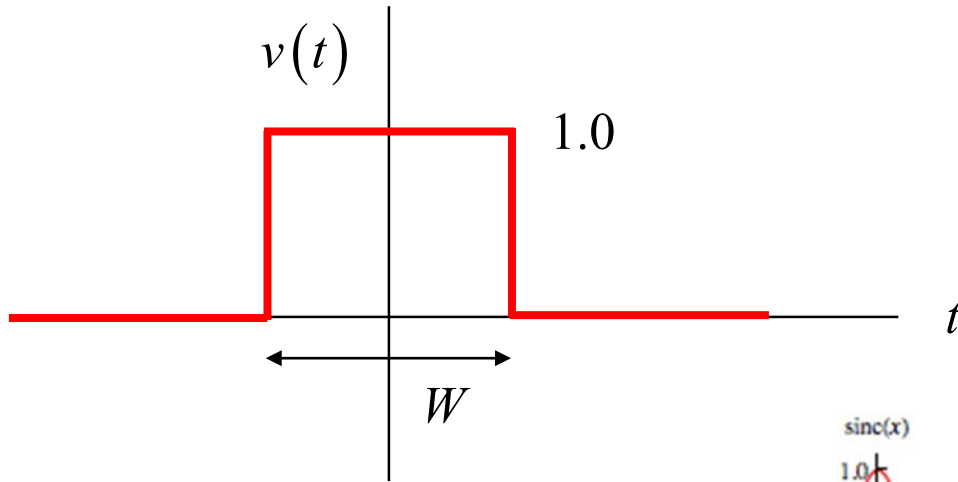
Phasor



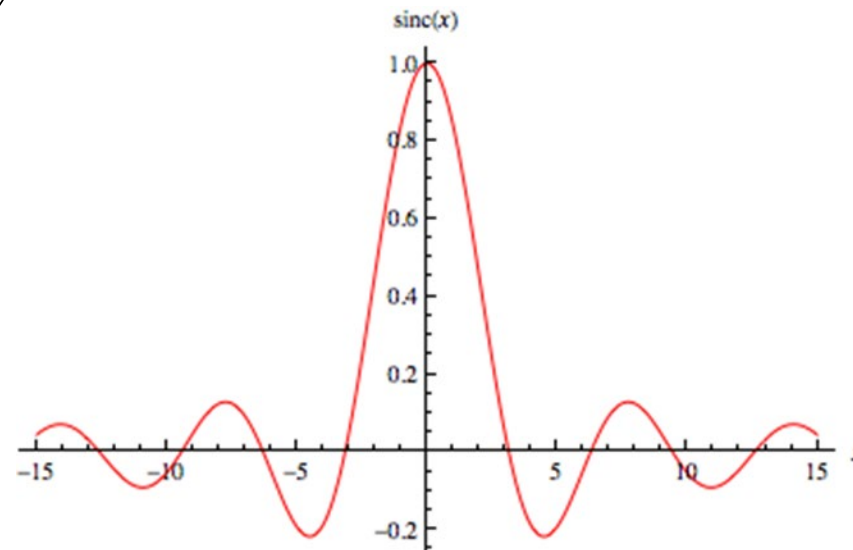
A pulse is resolved into a collection (spectrum) of infinite steady-state sinusoidal waves with different frequencies, amplitudes, and phases.

Frequency Domain (cont.)

Example: Rectangular pulse



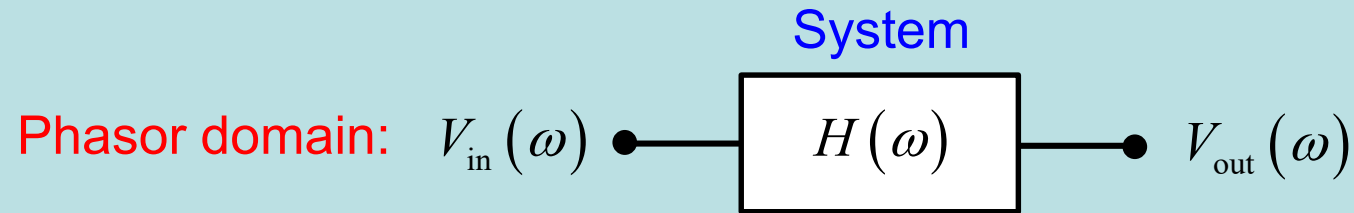
$$\tilde{v}(\omega) = \int_{-\infty}^{\infty} v(t) e^{-j\omega t} dt$$



$$\tilde{v}(\omega) = W \text{sinc}(W\omega / 2)$$

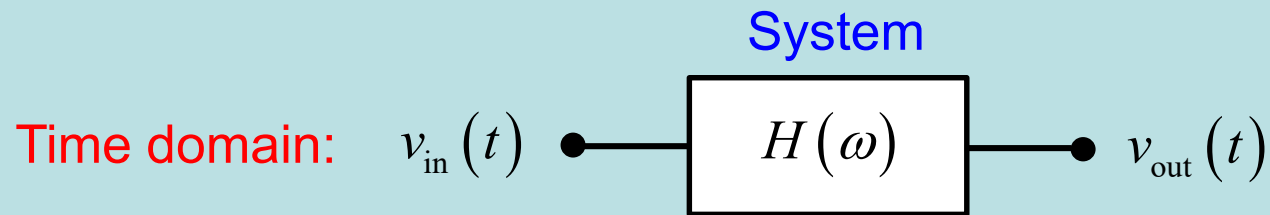
$$\text{sinc}(x) \equiv \frac{\sin(x)}{x}$$

Frequency Domain (cont.)



In the frequency domain, the system has a transfer function $H(\omega)$:

$$V_{\text{out}}(\omega) = H(\omega)V_{\text{in}}(\omega)$$



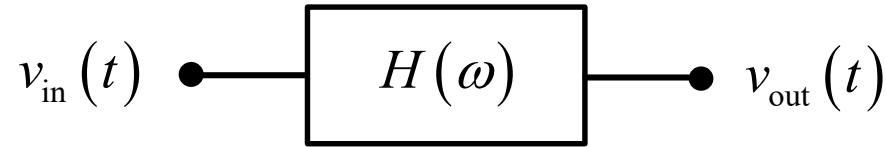
The time-domain response of the system to an input signal is:

$$v_{\text{out}}(t) = \frac{1}{\pi} \int_0^{\infty} \text{Re} \{ H(\omega) \tilde{v}_{\text{in}}(\omega) e^{j\omega t} \} d\omega$$

Frequency Domain (cont.)

Summary

System



Phasor domain :

$$H(\omega) \equiv \frac{V_{\text{out}}(\omega)}{V_{\text{in}}(\omega)}$$

$$v_{\text{out}}(t) = \frac{1}{\pi} \int_0^{\infty} \text{Re} \left\{ H(\omega) \tilde{v}_{\text{in}}(\omega) e^{j\omega t} \right\} d\omega$$

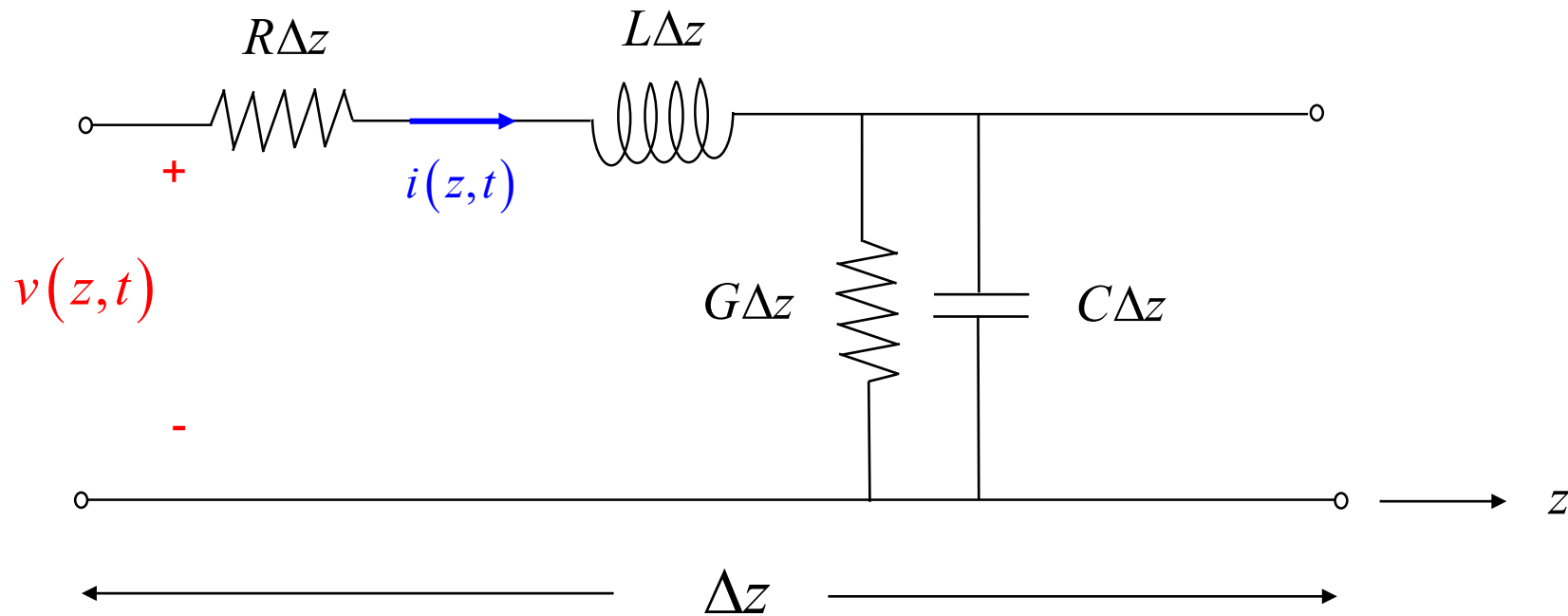
If we can solve the system in the phasor domain (i.e., get the transfer function $H(\omega)$), we can get the output for any time-varying input signal.

This is one reason why the phasor domain is so important!

This applies for transmission lines also!

Telegrapher's Equations

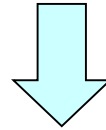
$$\frac{\partial^2 v}{\partial z^2} - (RG)v - (RC + LG)\frac{\partial v}{\partial t} - LC\left(\frac{\partial^2 v}{\partial t^2}\right) = 0$$



Frequency Domain

To convert to the phasor domain, we use: $\frac{\partial}{\partial t} \rightarrow j\omega$

$$\frac{\partial^2 v}{\partial z^2} - (RG)v - (RC + LG)\frac{\partial v}{\partial t} - LC\left(\frac{\partial^2 v}{\partial t^2}\right) = 0$$



$$\frac{\partial^2 V}{\partial z^2} - (RG)V - j\omega(RC + LG)V - (j\omega)^2 LC V = 0$$

or

$$\frac{d^2 V}{dz^2} = (RG)V + j\omega(RC + LG)V - (\omega^2 LC)V$$

Frequency Domain (cont.)

$$\frac{d^2V}{dz^2} = \left[(RG) + j\omega(RC + LG) - (\omega^2 LC) \right] V$$

Note that

$$RG + j\omega(RC + LG) - \omega^2 LC = (R + j\omega L)(G + j\omega C)$$

$$Z = R + j\omega L = \text{series impedance / length}$$

$$Y = G + j\omega C = \text{parallel admittance / length}$$

We can therefore write:

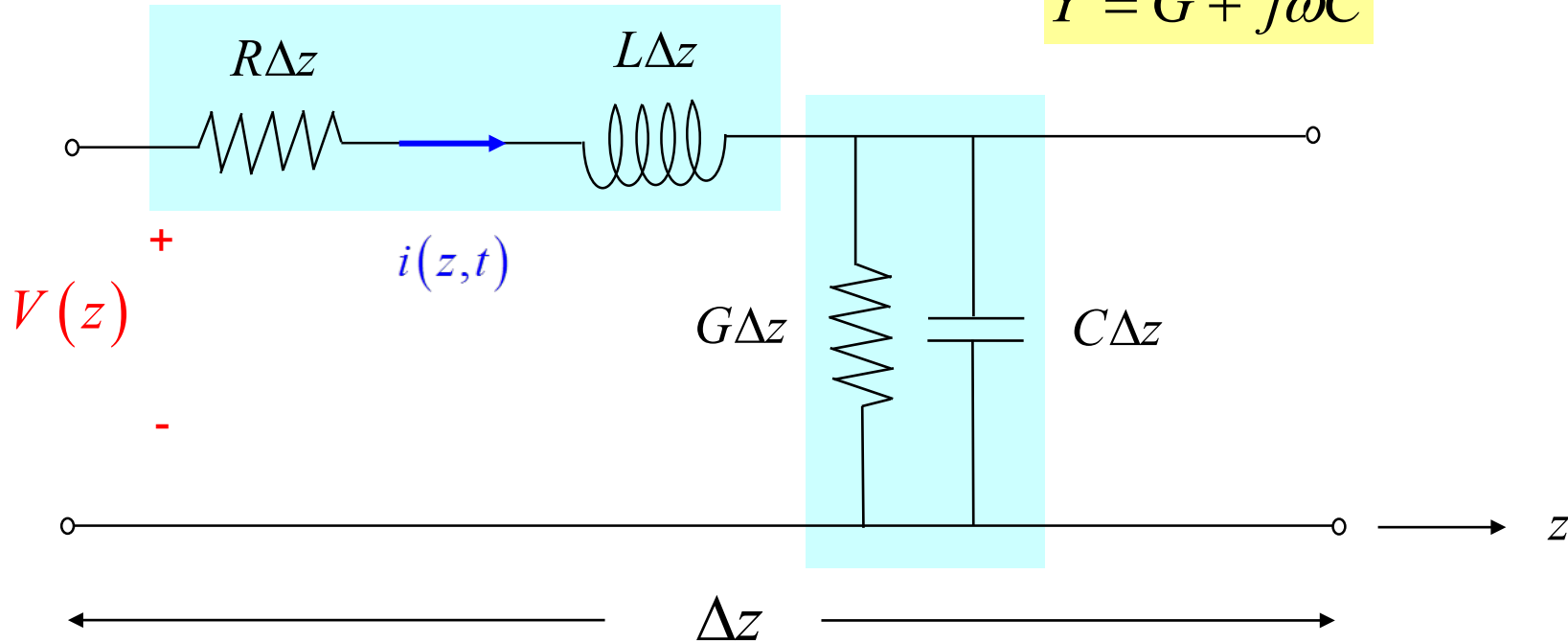
$$\frac{d^2V}{dz^2} = (ZY)V$$

Telegrapher's Equations

$$\frac{d^2 V}{dz^2} = (ZY)V$$

$$Z = R + j\omega L$$

$$Y = G + j\omega C$$



Frequency Domain (cont.)

$$\frac{d^2V}{dz^2} = (ZY)V$$

Define

$$\gamma^2 \equiv ZY = (R + j\omega L)(G + j\omega C)$$

Then

$$\frac{d^2V}{dz^2} = (\gamma^2)V$$

Solution:

$$V(z) = Ae^{-\gamma z} + Be^{+\gamma z}$$

Note: We have an exact solution, even for a lossy line, in the phasor domain!

Propagation Constant

Convention: We choose the (complex) square root to be the principal branch:

$$\gamma \equiv \sqrt{(R + j\omega L)(G + j\omega C)} \quad (\text{lossy case})$$

γ is called the **propagation constant**, with units of [1/m]

Review of principal branch of square root:

$$c = |c| e^{j\phi}$$

$$\sqrt{c} = \sqrt{|c|} e^{j(\phi/2)}$$

$$-\pi < \phi \leq \pi$$

Note:

$$-\pi / 2 < \phi / 2 \leq \pi / 2$$



$$\operatorname{Re} \sqrt{c} \geq 0$$

Propagation Constant (cont.)

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

Denote: $\gamma = \alpha + j\beta$

γ = propagation constant [1/m]
 α = attenuation constant [nepers/m]
 β = phase constant [radians/m]

Choosing the principal branch means that

$$\operatorname{Re} \gamma \geq 0 \quad \Rightarrow \quad \alpha \geq 0$$

Propagation Constant (cont.)

For a lossless line, we consider this as the limit of a lossy line, in the limit as the loss tends to zero:

$$\gamma = \sqrt{(\cancel{R} + j\omega L)(\cancel{G} + j\omega C)} = (\omega\sqrt{LC})\sqrt{-1}$$

Hence $\gamma = j\omega\sqrt{LC}$ (lossless case)

Hence, we have that

$$\alpha = 0$$

$$\beta = \omega\sqrt{LC}$$

Note: $\alpha = 0$ for a lossless line.

Propagation Constant (cont.)

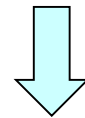
Physical interpretation of waves:

Note:

The waves must decay in the direction of propagation.

$$V^+(z) = A e^{-\gamma z} \quad (\text{forward traveling wave})$$

$$V^-(z) = B e^{+\gamma z} \quad (\text{backward traveling wave})$$



$$\gamma = \alpha + j\beta$$

Forward traveling wave: $V^+(z) = A e^{-\alpha z} e^{-j\beta z}$ (decaying as it travels)

Backward traveling wave: $V^-(z) = B e^{+\alpha z} e^{+j\beta z}$ (decaying as it travels)

Propagation Wavenumber

Alternative notations:

$$\gamma = \alpha + j\beta \quad (\text{propagation constant})$$

$$k_z = \beta - j\alpha \quad (\text{propagation wavenumber})$$

$$\text{Note: } \gamma = jk_z$$

$$V^+(z) = Ae^{-\gamma z} = Ae^{-jk_z z} = Ae^{-\alpha z} e^{-j\beta z}$$

Forward Wave

Forward traveling wave: $V^+(z) = A e^{-\alpha z} e^{-j\beta z}$

Denote $A = |A| e^{j\phi}$

Then $V^+(z) = |A| e^{j\phi} e^{-\alpha z} e^{-j\beta z}$

In the time domain we have:

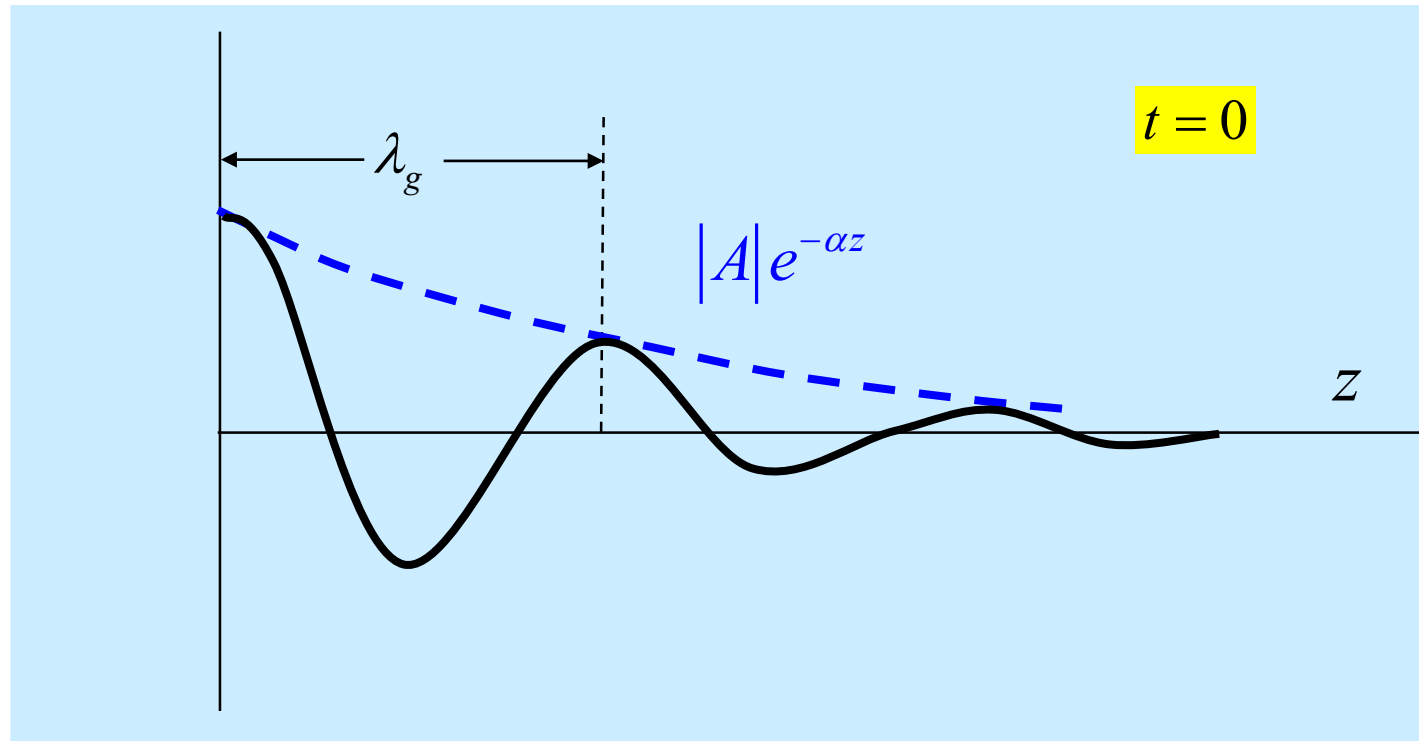
$$v^+(z, t) = \operatorname{Re} \left\{ V^+(z) e^{j\omega t} \right\}$$

Hence, we have $v^+(z, t) = |A| e^{-\alpha z} \cos(\omega t - \beta z + \phi)$

Forward Wave (cont.)

$$v^+(z, t) = |A| e^{-\alpha z} \cos(\omega t - \beta z + \phi)$$

Snapshot of Waveform:



The distance λ_g is the distance it takes for the waveform to “repeat” itself in meters.

$$\lambda_g = \text{guided wavelength}$$

Wavelength

The wave “repeats” (except for the amplitude decay) when:

$$\beta \lambda_g = 2\pi$$

Hence:

$$\beta = \frac{2\pi}{\lambda_g}$$

Note: This equation can be used to find λ_g if we already know β :

$$\lambda_g = \frac{2\pi}{\beta} = \frac{2\pi}{\text{Im}(\gamma)} = \frac{2\pi}{\text{Im}\left(\sqrt{(R + j\omega L)(G + j\omega C)}\right)}$$

Wavelength (cont.)

Lossless case:

$$\lambda_g = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{LC}} = \frac{2\pi}{2\pi f\sqrt{LC}} = \frac{1}{f\sqrt{LC}} = \frac{1}{f\sqrt{\mu\epsilon}} = \frac{c_d}{f} = \frac{c}{\sqrt{\mu_r\epsilon_r}} \frac{1}{f} = \frac{\lambda_0}{\sqrt{\mu_r\epsilon_r}} = \lambda_d$$

Summary for lossless case:

$$\lambda_g = \lambda_d$$

$$\lambda_d = \frac{\lambda_0}{\sqrt{\mu_r\epsilon_r}}$$

$$\lambda_0 = \frac{c}{f}$$

λ_d = wavelength in dielectric

λ_0 = wavelength in free - space (air)

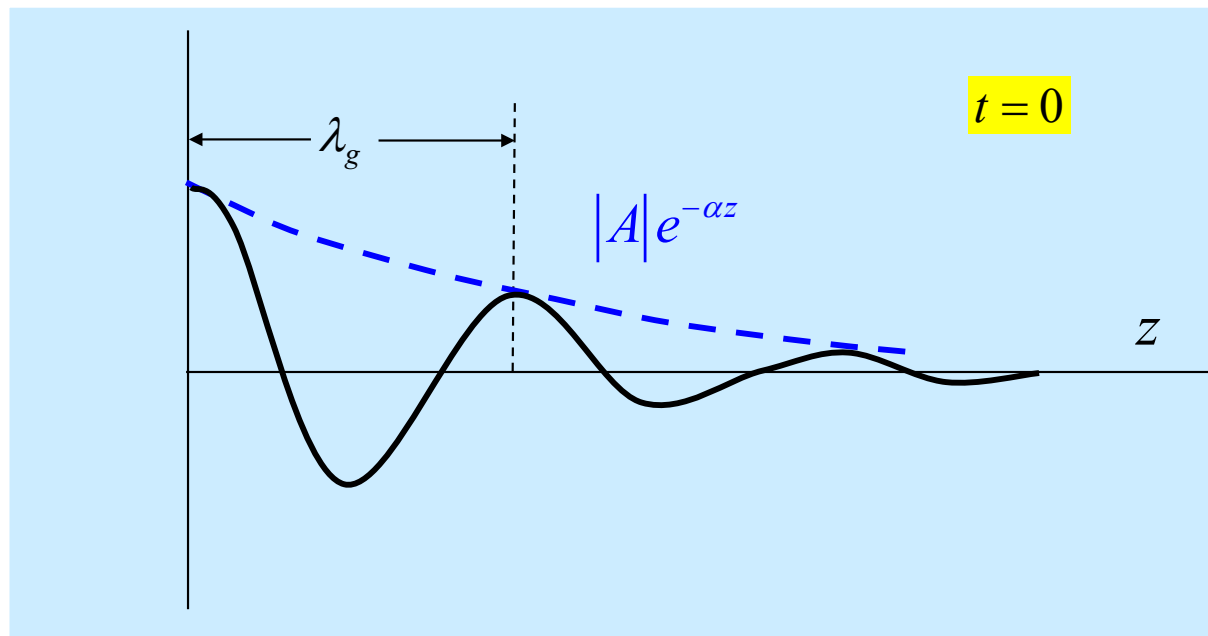
$$c \equiv 2.99792458 \times 10^8 \text{ [m/s]}$$

Attenuation Constant

The attenuation constant controls how fast the wave decays.

$$v^+(z, t) = |A| e^{-\alpha z} \cos(\omega t - \beta z + \phi)$$

$$\text{envelope} = |A| e^{-\alpha z}$$



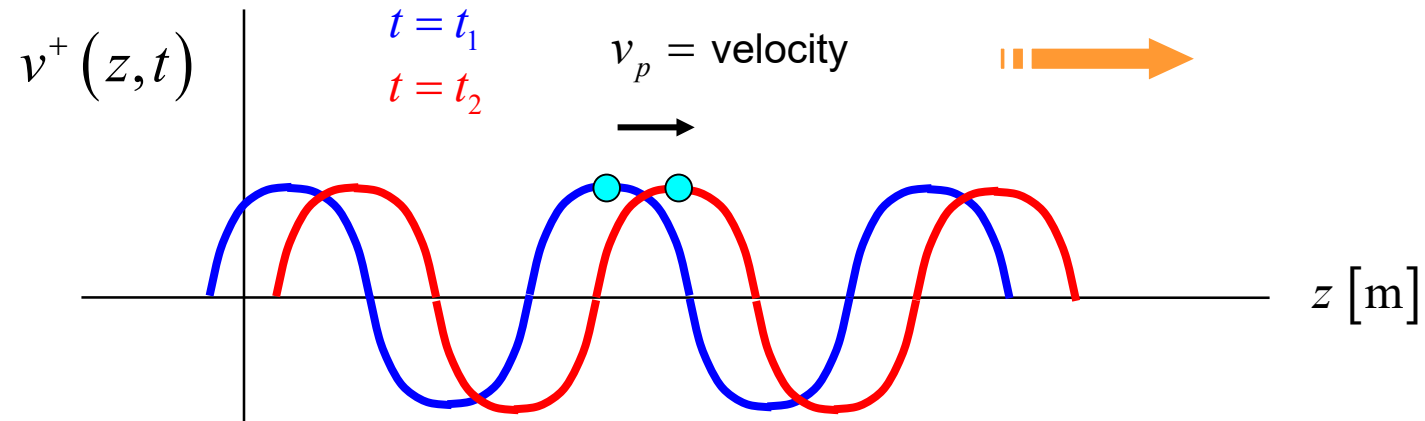
$$\alpha = \text{Re}(\gamma) = \text{Re}\left(\sqrt{(R + j\omega L)(G + j\omega C)}\right)$$

Phase Velocity

The forward-traveling wave is moving in the positive z direction.

Consider a sinusoidal wave moving on a transmission line
(shown in the figure below for a lossless line ($\alpha = 0$) for simplicity):

$$v^+(z, t) = |A| e^{-\alpha z} \cos(\omega t - \beta z + \phi)$$



Crest of wave: $\omega t - \beta z + \phi = 0$

Phase Velocity (cont.)

The **phase velocity** v_{phase} is the velocity of a point on the wave, such as the crest.

$$\text{Set } \omega t - \beta z = -\phi = \text{constant}$$

Take the derivative with respect to time: $\omega - \beta \frac{dz}{dt} = 0$

$$\text{Hence } \frac{dz}{dt} = \frac{\omega}{\beta}$$

We thus have

$$v_{\text{phase}} = \frac{\omega}{\beta}$$

Note:

This result holds for a general lossy line.

$$\text{Recall: } \beta = \text{Im}\left(\sqrt{(R + j\omega L)(G + j\omega C)}\right)$$

Phase Velocity (cont.)

Let's calculate the phase velocity for a lossless line:

$$v_{\text{phase}} = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{LC}} = \frac{1}{\sqrt{LC}}$$

Lossless line:

$$\alpha = 0$$

$$\beta = \omega\sqrt{LC}$$

Also, we know that $LC = \mu\varepsilon = \frac{1}{c_d^2}$

Hence $v_{\text{phase}} = c_d$ (lossless line)

Recall:

For a lossless line, any signal travels at the speed c_d .
(So this must be true for a sinusoidal signal!)

Recall: $c_d = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \frac{1}{\sqrt{\mu_r\varepsilon_r}} = \frac{c}{\sqrt{\mu_r\varepsilon_r}}$

Backward Traveling Wave

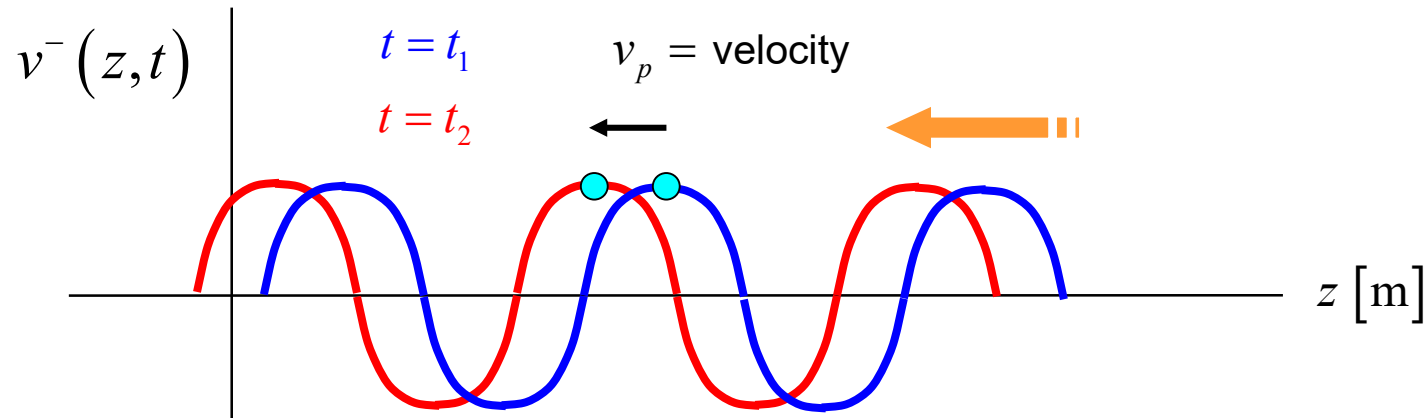
Let's now consider the backward-traveling wave

(shown in the figure below for a lossless line ($\alpha = 0$) for simplicity):

$$V^-(z) = Be^{+\gamma z} = B e^{+\alpha z} e^{+j\beta z} = |B| e^{j\psi} e^{+\alpha z} e^{+j\beta z}$$

Denote $B = |B| e^{j\psi}$

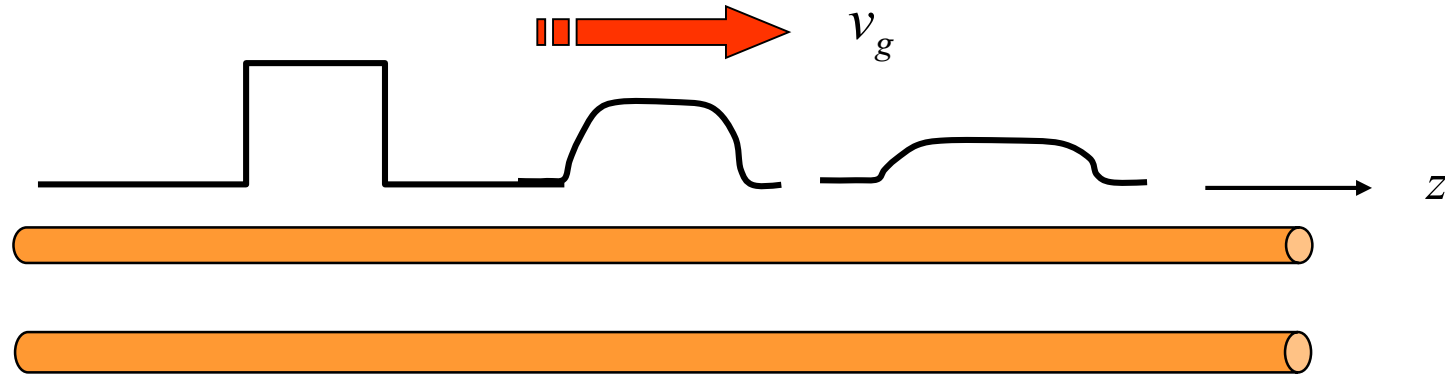
⇒ $v^-(z, t) = |B| e^{+\alpha z} \cos(\omega t + \beta z + \psi)$



This wave has the same phase velocity, but it travels backwards.

Group Velocity

The group velocity v_{group} is the velocity of a pulse.



We have (derivation omitted):

$$v_{\text{group}} = \frac{d\omega}{d\beta}$$

Note:

For a lossy line, the pulse will be distorting as it propagates down the line.

Note: for a lossless line we have:

$$v_{\text{group}} = v_{\text{phase}} = 1 / \sqrt{LC} = 1 / \sqrt{\mu\epsilon} = c_d$$

(Lossless line: $\beta = \omega\sqrt{LC} \Rightarrow d\omega / d\beta = 1 / \sqrt{LC}$)

Attenuation in dB/m

$$V^+(z) = A e^{-\gamma z} = A e^{-jk_z z} = A e^{-\alpha z} e^{-j\beta z}$$

Gain in dB: $\text{dB} = 20 \log_{10} \left| \frac{V^+(z)}{V^+(0)} \right| = 20 \log_{10} (e^{-\alpha z})$

Use the following logarithm identity: $\log_{10} x = \frac{\ln x}{\ln 10}$

Therefore, the “gain” is: $\text{dB} = 20 \frac{\ln(e^{-\alpha z})}{\ln 10} = 20 \frac{(-\alpha z)}{\ln 10} = -\left(\frac{20}{\ln 10} \alpha \right) z$

Hence, we have: $\text{Attenuation} = \left(\frac{20}{\ln 10} \right) \alpha \quad [\text{dB/m}]$

Attenuation in dB/m (cont.)

Final attenuation formulas:

$$\text{Attenuation} = \left(\frac{20}{\ln 10} \right) \alpha \quad [\text{dB/m}]$$

$$\text{Attenuation} \approx (8.686) \alpha \quad [\text{dB/m}]$$

Example: Coaxial Cable

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)} \quad [\text{F/m}]$$

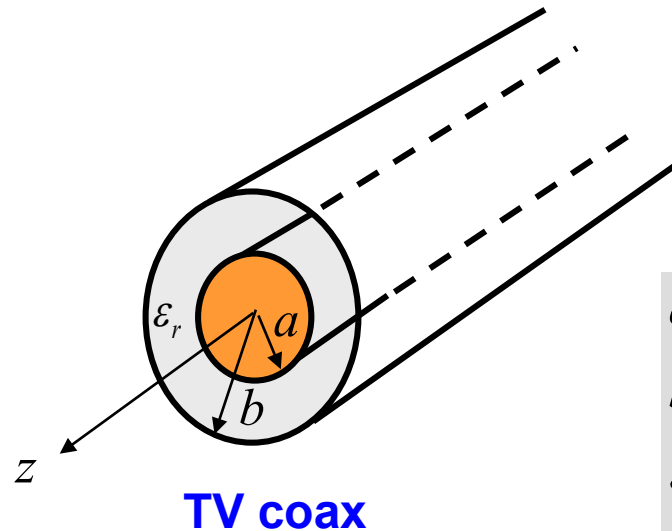
$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad [\text{H/m}]$$

$$G = \frac{2\pi\sigma_d}{\ln\left(\frac{b}{a}\right)} \quad [\text{S/m}]$$

$$R = \left(\frac{1}{2\pi a \sigma_{ma} \delta_{ma}} + \frac{1}{2\pi b \sigma_{mb} \delta_{mb}} \right) \quad [\Omega/\text{m}]$$

$$\delta_{ma} = \sqrt{\frac{2}{\omega\mu_{ma}\sigma_{ma}}} \quad \delta_{mb} = \sqrt{\frac{2}{\omega\mu_{mb}\sigma_{mb}}}$$

(skin depth of the two conductors)



Copper conductors
(nonmagnetic: $\mu_m = \mu_0$)

$$a = 0.5 \text{ [mm]}$$

$$b = 3.2 \text{ [mm]}$$

$$\epsilon_r = 2.2$$

$$\tan \delta_d = 0.001$$

$$\sigma_{ma} = \sigma_{mb} = 5.8 \times 10^7 \text{ [S/m]}$$

$$f = 500 \text{ [MHz] (UHF)}$$

Note:

The “loss tangent” of the dielectric is called $\tan \delta_d$.

$$\tan \delta_d \equiv \frac{\sigma_d}{\omega\epsilon} = \frac{\sigma_d}{\omega\epsilon_0\epsilon_r}$$

Example: Coaxial Cable (cont.)

Characteristic impedance (ignore R and G for this):

$$Z_0 = \sqrt{\frac{L}{C}} = \frac{\eta_0}{2\pi\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right)$$

$$Z_0 = 75 \text{ } [\Omega]$$

Skin depth of metal:

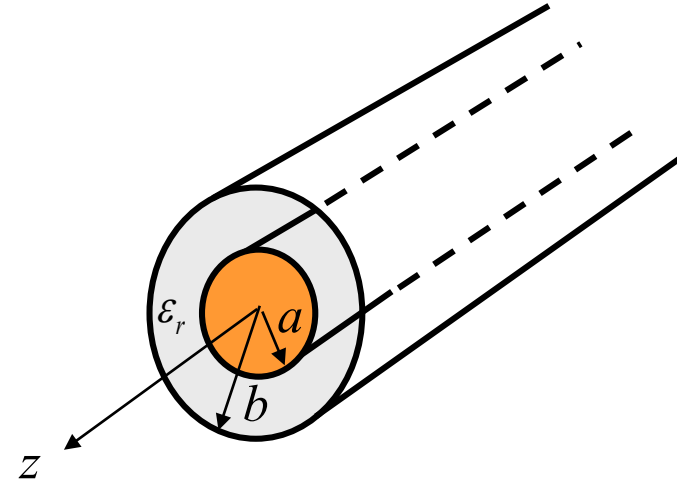
$$\delta_m = \sqrt{\frac{2}{\omega\mu\sigma_m}} \quad (\mu = \mu_0)$$

$$\delta_m = 2.955 \times 10^{-6} \text{ } [\text{m}]$$

Effective conductivity of dielectric:

$$\sigma_d = (\omega\epsilon_0\epsilon_r) \tan \delta_d$$

$$\sigma_d = 6.12 \times 10^{-5} \text{ } [\text{S/m}]$$



$$a = 0.5 \text{ } [\text{mm}]$$

$$b = 3.2 \text{ } [\text{mm}]$$

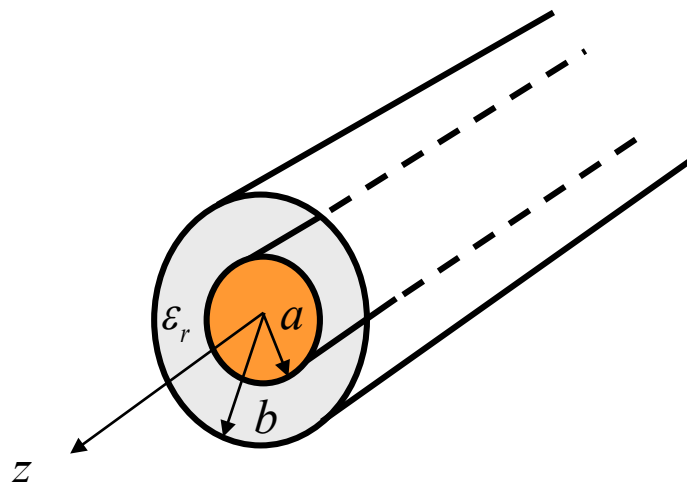
$$\epsilon_r = 2.2$$

$$\tan \delta_d = 0.001$$

$$\sigma_{ma} = \sigma_{mb} = 5.8 \times 10^7 \text{ } [\text{S/m}]$$

$$f = 500 \text{ } [\text{MHz}] \text{ (UHF)}$$

Example: Coaxial Cable (cont.)



$$a = 0.5 \text{ [mm]}$$

$$b = 3.2 \text{ [mm]}$$

$$\epsilon_r = 2.2$$

$$\tan \delta_d = 0.001$$

$$\sigma_{ma} = \sigma_{mb} = 5.8 \times 10^7 \text{ [S/m]}$$

$$f = 500 \text{ [MHz] (UHF)}$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

Results for (R, L, G, C) :

$$R = 2.147 \text{ } [\Omega/\text{m}]$$

$$L = 3.713 \times 10^{-7} \text{ [H/m]}$$

$$G = 2.071 \times 10^{-4} \text{ [S/m]}$$

$$C = 6.593 \times 10^{-11} \text{ [F/m]}$$

$$\gamma = 0.022 + j(15.543) \text{ [1/m]}$$

$$\alpha = 0.022 \text{ [nepers/m]}$$

$$\beta = 15.544 \text{ [rad/m]}$$

$$\text{Attenuation} = 0.191 \text{ [dB/m]}$$

$$\lambda_g = 0.404 \text{ [m]}$$

Current

Use the first Telegrapher equation:

$$\frac{\partial v}{\partial z} = -Ri - L \frac{\partial i}{\partial t}$$

$$\Downarrow \quad \frac{\partial}{\partial t} \rightarrow j\omega$$

$$\frac{\partial V}{\partial z} = -RI - j\omega LI$$

Next, use

$$V(z) = Ae^{-\gamma z} + Be^{+\gamma z}$$

so
$$\frac{\partial V(z)}{\partial z} = -\gamma \left[Ae^{-\gamma z} - Be^{+\gamma z} \right]$$

Current (cont.)

Hence, we have:

$$-\gamma \left[A e^{-\gamma z} - B e^{+\gamma z} \right] = -RI - j\omega LI$$

Solving for the phasor current I , we have:

$$\begin{aligned} I &= \left(\frac{\gamma}{R + j\omega L} \right) \left[A e^{-\gamma z} - B e^{+\gamma z} \right] \\ &= \left(\frac{\sqrt{(R + j\omega L)(G + j\omega C)}}{R + j\omega L} \right) \left[A e^{-\gamma z} - B e^{+\gamma z} \right] \\ &= \sqrt{\frac{G + j\omega C}{R + j\omega L}} \left[A e^{-\gamma z} - B e^{+\gamma z} \right] \end{aligned}$$

Characteristic Impedance

Define the (complex) characteristic impedance Z_0 in the frequency domain for a lossy line:

$$Z_0 \equiv \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

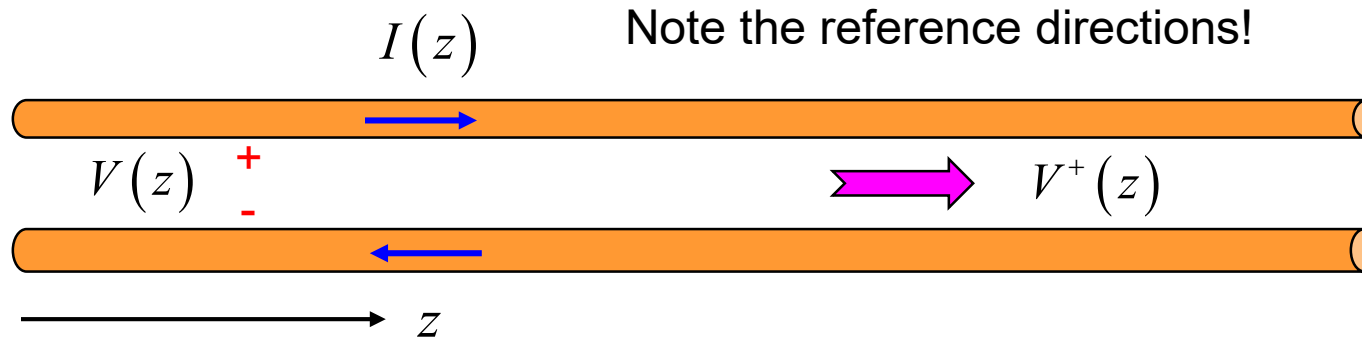
Then we have:

$$I(z) = \left(\frac{1}{Z_0} \right) \left[A e^{-\gamma z} - B e^{+\gamma z} \right]$$

Note:

In the time domain, we only defined Z_0 for a lossless line. In the frequency domain, we can define it for a lossy line.

Characteristic Impedance (cont.)



$$\frac{V^+(z)}{I^+(z)} = Z_0$$

The characteristic impedance is the ratio of the voltage to the current, for a wave traveling in the positive z direction.

Practical note: Even though Z_0 is always complex for a practical line (due to loss), we usually neglect this and take it to be real.

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \approx \sqrt{\frac{L}{C}}$$

Summary of Solution

Characteristic Impedance

Lossy

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Lossless:

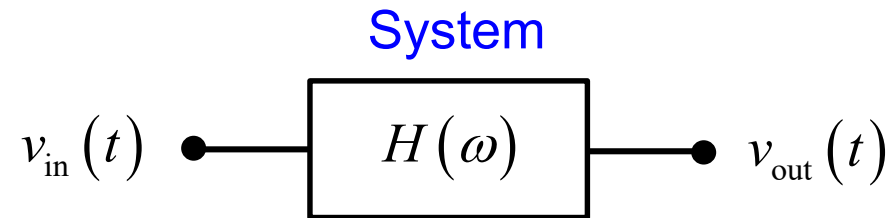
$$Z_0 = \sqrt{\frac{L}{C}}$$

Voltage and Current

$$V(z) = Ae^{-\gamma z} + Be^{+\gamma z}$$

$$I(z) = \left(\frac{1}{Z_0} \right) [Ae^{-\gamma z} - Be^{+\gamma z}]$$

Numerical Solution for Lossy Line



Phasor domain:

$$H(\omega) \equiv \frac{V_{\text{out}}(\omega)}{V_{\text{in}}(\omega)}$$

$$v_{\text{out}}(t) = \frac{1}{\pi} \int_0^{\infty} \text{Re} \left\{ H(\omega) \tilde{v}_{\text{in}}(\omega) e^{j\omega t} \right\} d\omega$$

This must be evaluated numerically
(e.g., using MATLAB).

Lossy transmission line:

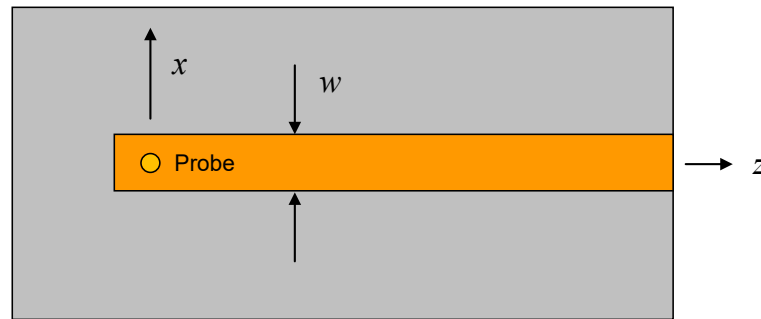


$$H(\omega) = e^{-\gamma z_0} = e^{-z_0 \sqrt{(R+j\omega L)(G+j\omega C)}}$$

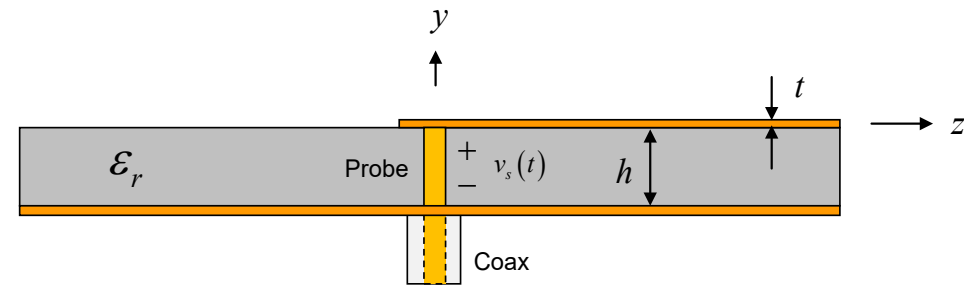
Numerical Solution for Lossy Line (cont.)

Example: Propagation on a lossy microstrip line

(From ECE 5317)



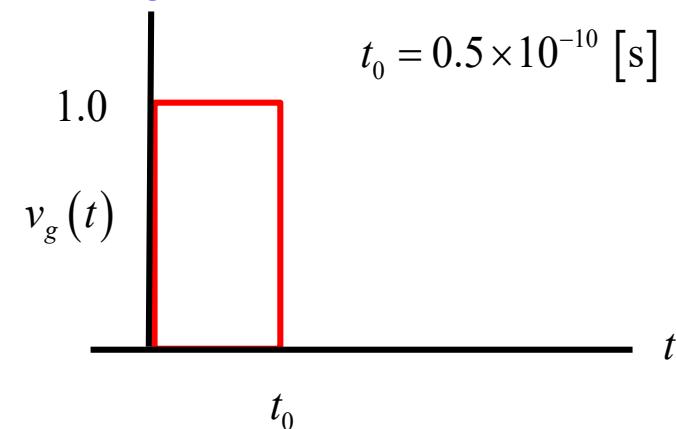
TOP VIEW



SIDE VIEW

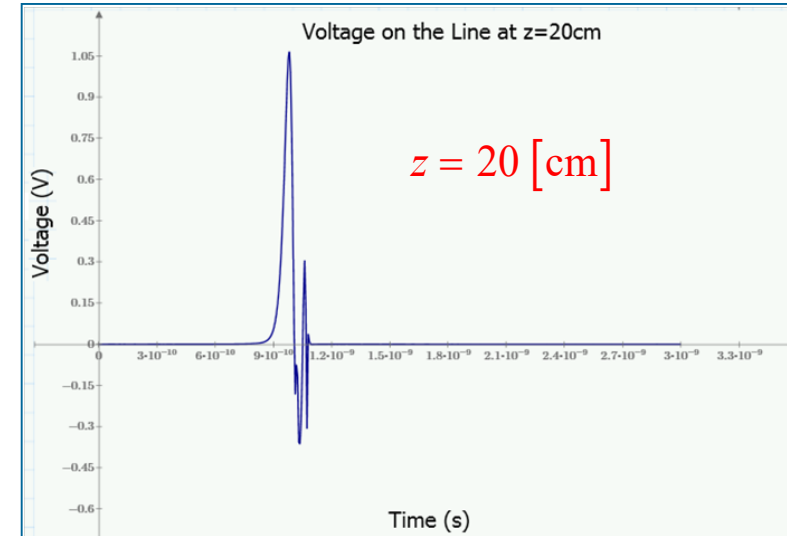
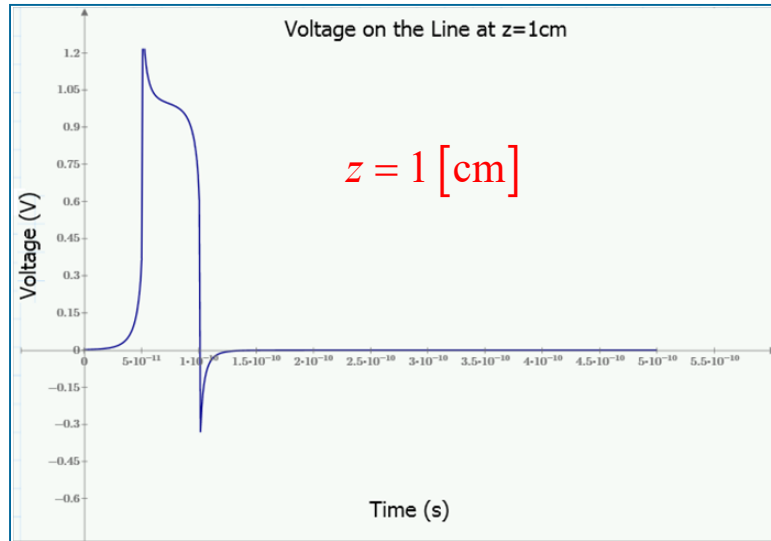
$\epsilon_r = 2.33$
 $\tan \delta = 0.001$
 $h = 0.787 \text{ [mm]} \text{ (31 mils)}$
 $w = 2.35 \text{ [mm]}$
 $t = 0.0175 \text{ [mm]} \text{ ("half oz" copper cladding)}$
 $\sigma_m = 3.0 \times 10^7 \text{ [S/m]}$

Input signal:



Numerical Solution for Lossy Line (cont.)

Example: Propagation on a microstrip line



$$\epsilon_r = 2.33$$

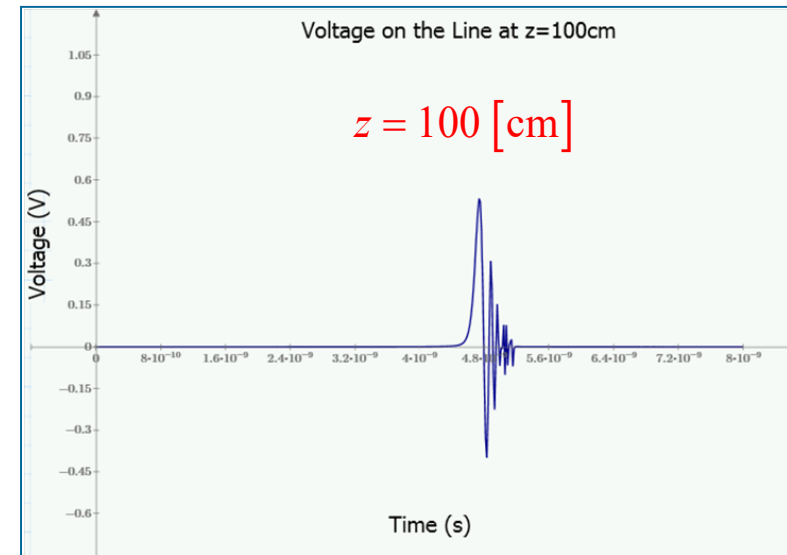
$$\tan \delta = 0.001$$

$$h = 0.787 \text{ [mm]} \text{ (31 mils)}$$

$$w = 2.35 \text{ [mm]}$$

$$t = 0.0175 \text{ [mm]} \text{ ("half oz" copper cladding)}$$

$$\sigma_m = 3.0 \times 10^7 \text{ [S/m]}$$



Appendix: Summary of Formulas

General Lossy Case

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$\tan \delta_d \equiv \frac{\sigma_d}{\omega \epsilon} = \frac{\sigma_d}{\omega \epsilon_0 \epsilon_r}$$

$$V(z) = A e^{-\gamma z} + B e^{+\gamma z}$$

$$\frac{G}{\omega C} = \tan \delta_d$$

$$I(z) = \left(\frac{1}{Z_0} \right) \left[A e^{-\gamma z} - B e^{+\gamma z} \right]$$

$$\beta = \frac{2\pi}{\lambda_g}$$

$$\gamma \equiv \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$v_{\text{phase}} = \frac{\omega}{\beta}$$

$$\gamma = \alpha + j\beta$$

$$\text{Attenuation} = (8.686) \alpha \quad [\text{dB/m}]$$

Appendix: Summary of Formulas

Lossless Case

$$Z_0 = \sqrt{\frac{L}{C}}$$

$$V(z) = Ae^{-j\beta z} + Be^{+j\beta z}$$

$$I(z) = \left(\frac{1}{Z_0}\right) [Ae^{-j\beta z} - Be^{+j\beta z}]$$

$$\gamma = j\beta$$

$$\beta = \omega\sqrt{LC}$$

$$c = 2.99792458 \times 10^8 \text{ [m/s]}$$

$$\lambda_g = \lambda_d$$

$$\beta = \frac{2\pi}{\lambda_d}$$

$$\lambda_d = \frac{\lambda_0}{\sqrt{\epsilon_r \mu_r}}$$

$$\lambda_0 = \frac{c}{f}$$

$$v_{\text{phase}} = c_d$$

$$c_d = \frac{c}{\sqrt{\epsilon_r \mu_r}}$$